IS THERE SIGNAL IN THE VASCULAR NETWORKS?

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Abstract

- Does the vessel network around a breast tumour predict response to chemotherapy? We ask that in two separable parts.
- **Vessels alone beat chance:** pooled out-of-fold AUC 0.62 $[0.54, 0.70]$, $N = 179$, no tumour mask or clinical variables.
- **The graph adds nothing yet:** paired ΔAUC $+0.002$ $[-0.041, +0.046]$ vs. a tabular summary of the same features.

Introduction

1. Background. Patients reaching **pCR** — pathologic complete response, no residual invasive cancer at surgery — do markedly better. **DCE-MRI** images the breast repeatedly while a contrast agent washes through it, recording how each vessel perfuses.

2. Research question.

Is there signal in the vascular networks?

1. **Is there signal in the blood vessels at all?**
2. **Does the graph add anything beyond that?**

Methods: scan to graph

Graph representations

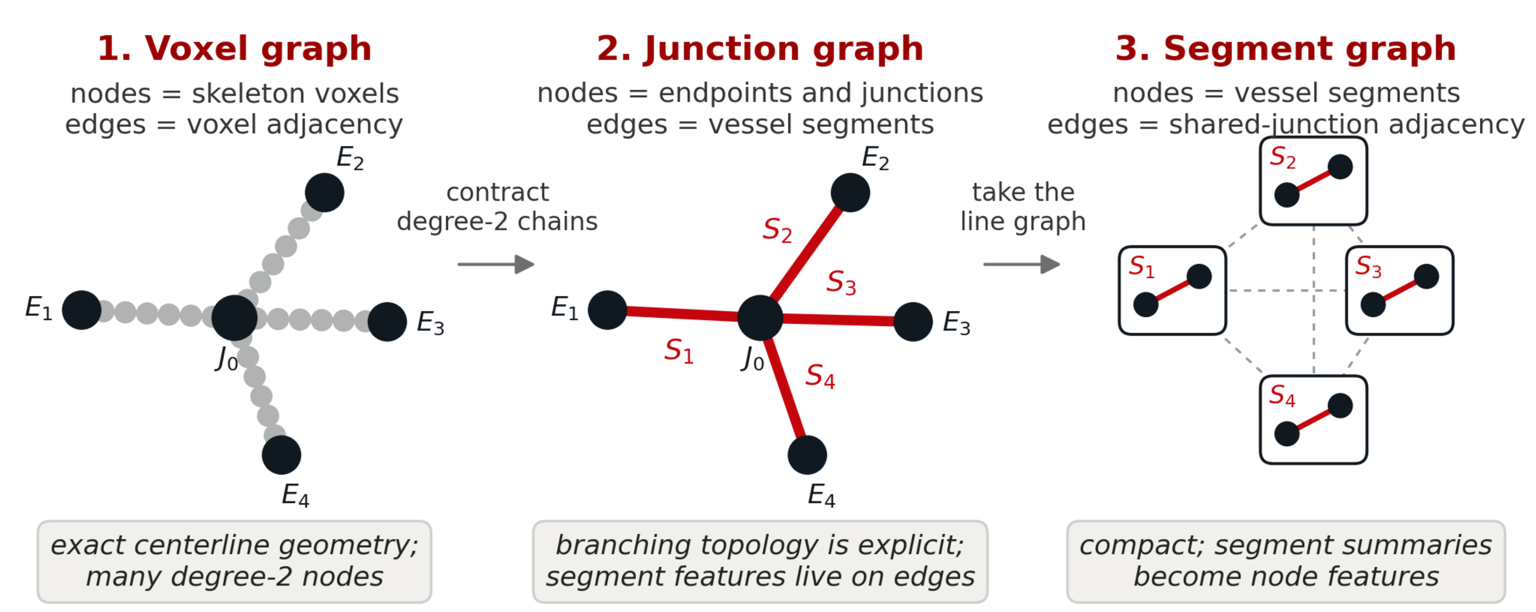

Fig. 2: Three deterministic transforms of the same skeleton, compared on identical folds.

Node features (7). Peak time • peak enhancement • time to enhancement • wash-in slope • wash-out slope • positive AUC • vessel radius. **Edge features** summarise each segment: length, tortuosity, volume, radius and curvature statistics, mean kinetics along it.

Contrast-forecasting pretraining

A new contribution of this project.

What is thrown away above: each node's 24-point enhancement curve becomes six scalars before the model sees it, and edges carry no flow direction, so the graph can only average static kinetics. The models never see contrast *move*. So we pretrain the encoder on a label-free task: **given the first frames, predict every node's enhancement over the next frames**, using message passing — so the encoder has to learn how the bolus propagates along vessels. After Tang

1 In the vessels? YES

UChicago ultrafast DCE-MRI, $N = 179$, **vessels only** — no tumour mask, no clinical variables. Pooled out-of-fold AUC, case-resampling bootstrap CIs.

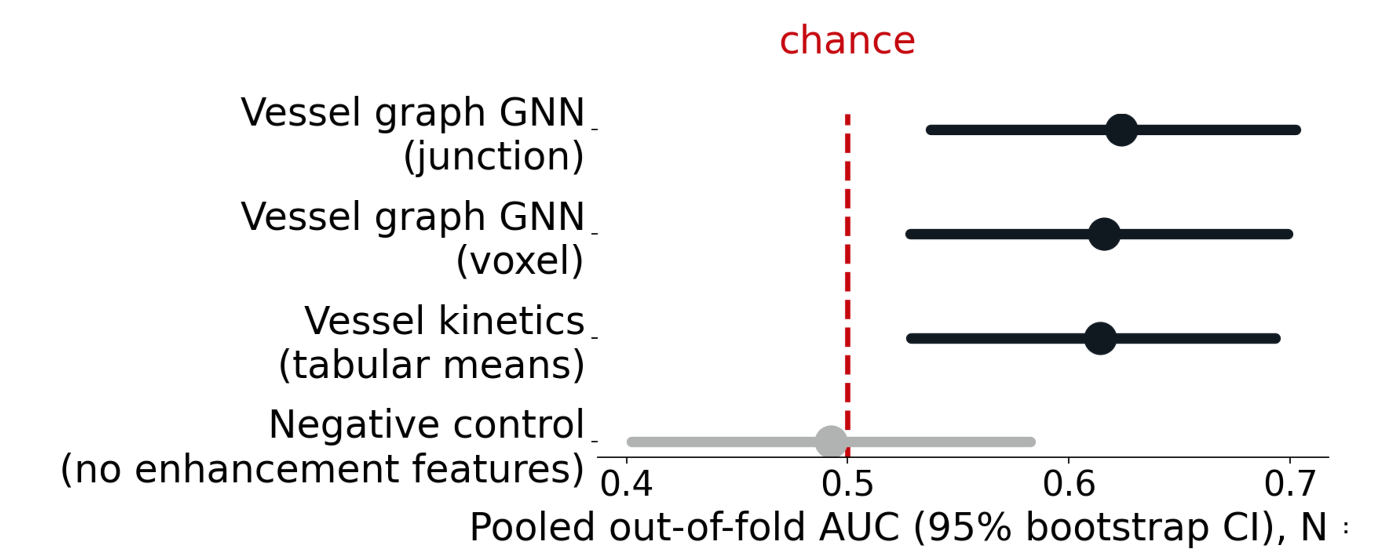

Fig. 4: Every vessel model clears chance; the control does not.

Bootstrap lower bounds clear 0.5. A control built from the only two node features carrying *no* enhancement information — peak time and radius — sits at chance, so the signal is specifically in the contrast kinetics.

Independently, on I-SPY2 [1] ($n = 808$), adding vessel features to clinical variables and tumour size raises AUC from 0.571 ± 0.046 to 0.606 ± 0.028 (XGBoost); the paired gain for logistic regression is $+0.024$, $p = 0.025$.

2 From the graph? NOT YET